SUMUL Marketing Dashboard - Project Report

Marketing Department, SUMUL Dairy, Surat · Oct 14, 2026 · @sapna

Cover Page

Comprehensive Consumer Analytics & Market Intelligence Dashboard for SUMUL Dairy

A project report on the SUMUL customer survey, prepared for the Marketing Department, SUMUL Dairy, Surat.

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Declaration

We declare that this report is our own work. The analysis, data model, DAX measures and dashboard described here were built by us from the SUMUL customer survey data, and all outside sources are listed in the References.

Acknowledgement

We thank the Marketing Department of SUMUL Dairy for access to the survey data and for guidance on how the department uses customer feedback. We are grateful to Marketing Department for direction throughout the project.

Executive Summary

The dashboard shows that Sumul brand customer satisfaction (CSAT) is 86.1%, which is 3.9 points below the 90% target and below the 91.7% CSAT for the milk category as a whole.

The analysis covers 7,350 survey responses from 4,028 distinct customers across the Surat zones. Pouch milk is the main format (72.4% of customers), and Sumul/Amul holds 77.8% of pouch preference against 11.5% for Mahi. Availability and freshness are the top satisfaction drivers, together 45% of primary choices.

Quality is rated well (3.17 out of 4), but promotional schemes score only 1.86 and price value 2.59. Among 13.92% negative feedback, price and schemes (354 responses) and packaging quality (345) are the largest dissatisfaction themes. Probiotic dahi awareness is low at 38.31%.

The report documents the full workflow: data cleaning in Power Query, a star-schema model, DAX measures, six dashboard pages and six recommendations for the Marketing Department.

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Chapter 1: Introduction

1.1 Introduction to the Project

This project turns the SUMUL customer survey into an interactive Power BI dashboard that the Marketing Department can use to track satisfaction, brand preference and customer feedback in Surat. The survey holds 7,350 responses from 4,028 distinct customers.

The work follows a fixed path from raw survey data to business recommendations, shown in Figure 1.1: clean the data, model it, write measures, analyse the survey and publish the dashboard.

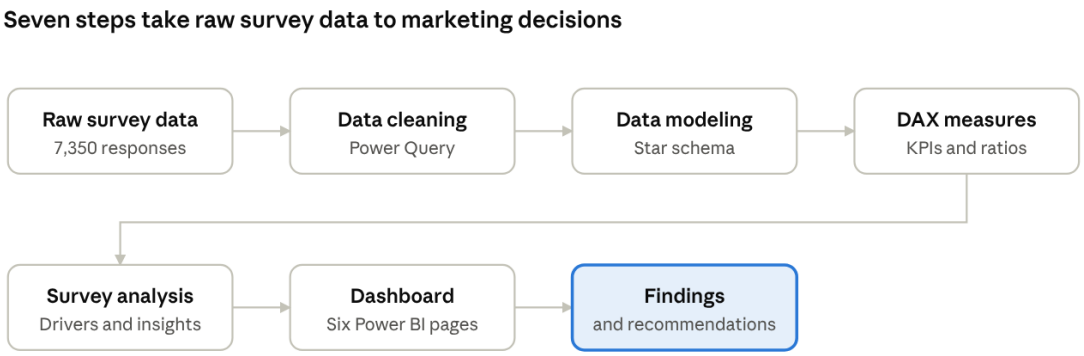


Figure 1.1: Project workflow from raw survey data to recommendations.

1.2 Overview of SUMUL

SUMUL (The Surat District Co-operative Milk Producers' Union Ltd.) is a pillar of the dairy co-operative movement in South Gujarat. It processes milk collected from thousands of village-level co-operative societies and sells pouch milk and milk products under the Sumul and Amul brands. The survey covers pouch milk, dahi, ghee, paneer, lassi, shrikhand, flavoured milk and bread, and the pack variants Amul Gold, Amul Taaza, Amul Slim & Trim and Sumul Cow Milk.

.3 Introduction to the Marketing Department

The Marketing Department decides how SUMUL products are priced, promoted, packed and delivered. It needs to know which customers are satisfied, why they choose a brand, which zones lag and what customers complain about. This dashboard gives the department those answers from one source.

1.4 Need for a Marketing Dashboard

Survey results held in spreadsheets are slow to read and hard to compare across zones, segments and brands. A dashboard answers three needs:

- Monitor satisfaction against the 90% target month by month.
- Compare SUMUL with competing brands such as Mahi, Gokulam and Rajhansh.
- Find the zones, products and complaints that need action first.

1.5 Problem Statement

SUMUL collects detailed customer feedback but has no single view that links satisfaction, buying behaviour, brand choice and complaints. Without it, the Marketing Department cannot easily see why Sumul brand CSAT (86.1%) sits below the category (91.7%) or where to act.

1.6 Project Objectives

1. Clean and validate the survey data so that every figure in the dashboard can be trusted.
2. Build a star-schema data model in Power BI for fast, consistent filtering.
3. Create DAX measures for satisfaction, market share, quality scores, awareness and sentiment.
4. Analyse survey responses by zone, age group, customer segment and brand.
5. Design a six-page interactive dashboard for the Marketing Department.
6. Recommend marketing actions based on the evidence.

1.7 Scope of the Project

Area	In scope
Data	One customer survey of 7,350 responses
Geography	Surat zones in the survey, such as Adajan, Vesu, Athwa and Varachha
Products	Pouch milk, dahi, ghee, paneer, lassi, shrikhand, flavoured milk, bread
Output	Six-page Power BI dashboard and this report

Chapter 2: Tools and Technologies

2.1 Introduction to Microsoft Power BI

Power BI is Microsoft's business intelligence tool. It connects to data, transforms it, models it and publishes interactive reports. This project uses Power BI Desktop for building and the Power BI Service for sharing.

2.2 Power Query for Data Transformation

Power Query is the data preparation layer inside Power BI. Each cleaning action is recorded as an applied step, so the whole process can be repeated when new survey data arrives. It was used to fix data types, remove duplicates, handle blanks and standardise names (Chapter 3).

2.3 Data Modeling and Star Schema

A star schema keeps fact tables of survey results and several small dimension tables, such as customer, date, location and product, around them. This layout gives simple one-to-many relationships, quick filtering and fewer calculation errors than a single wide table (Chapter 4).

2.4 DAX (Data Analysis Expressions)

DAX is the formula language of Power BI. A measure is calculated at the moment a report is filtered, so the same measure gives the right answer for every zone, month or segment selected. Examples are Sumul Brand CSAT % and Pouch Milk Penetration % (Chapter 5).

2.5 Data Visualization Techniques

The dashboard uses visuals matched to the question being asked:

- KPI cards for headline numbers.
- A gauge against the 90% target for satisfaction.
- Bar and column charts to rank brands, zones and drivers.
- Doughnut charts for shares such as milk format and pack variant.
- A line chart for the monthly satisfaction trend.
- Stacked bars for percentages that add to 100, such as brand choice drivers.
- Matrix and table visuals for spend bands and the feedback explorer.
- Slicers on every page for month, zone, segment, brand and similar filters.

2.6 Software and Tools Used

Tool	Purpose in this project
Microsoft Power BI Desktop	Data cleaning, modelling, DAX and dashboard design
Power Query Editor	Data transformation
DAX	Measures and KPIs
Microsoft Excel	Survey data source and spot checks
Power BI Service	Publishing and sharing

Tool	Purpose in this project
Microsoft Word	Report

Chapter 3: Data Understanding and Cleaning

3.1 Overview of the Marketing Dataset

The dataset is a large consumer survey of SUMUL customers in Surat: 7,350 completed responses from 4,028 distinct customers over 365 calendar days. Each response records the customer's milk consumption, spend, brand choice, ratings of five service attributes, price perception of seven dairy products, bread and dahi habits, and written feedback.

3.2 Data Sources and Dataset Structure

The survey was cleaned and reshaped into seven star-schema CSV files, which Power BI reads through Power Query. Table 3.1 lists each file, its size and its role.

Table	Type	Rows	Columns	What one row represents
FactSurveyResponse	Fact	7,350	42	One completed respondent survey
FactAttributeRating	Fact	36,750	9	One rated attribute for one survey (5 per survey)
FactProductPricePerception	Fact	51,450	8	One product price rating for one survey (7 per survey)
DimCustomer	Dimension	4,028	10	One surveyed customer
DimDate	Dimension	365	15	One calendar day
DimLocation	Dimension	12	8	One Surat zone
DimProduct	Dimension	10	9	One product or SKU

3.3 Identification of Data Quality Issues

The raw survey answers were unstructured: long question text as column names, mixed text and numbers, blanks, stray spaces and ratings stored as words. Table 3.2 lists each issue and how the Power Query pipeline fixed it.

Issue in the raw data	Treatment
Long survey questions used as column headers	Renamed to structured business names, for example MilkTypeConsumed and DeliveryChannel
Dates stored as text timestamps	Parsed to calendar dates and keyed as DateKey in YYYYMMDD form
Numbers stored as text (litres, age, spend)	Cast to decimal and whole-number types
Leading and trailing spaces in typed answers	Cleaned with Text.Trim and Text.Clean
Blank or non-applicable answers	Replaced with explicit placeholders such as "Not Applicable" and "No Specific Suggestion"
Ratings stored as words	Mapped to a 1 to 4 scale: Excellent = 4, Good = 3, Average = 2, Poor = 1
One wide table mixing many questions	Split into three fact tables and four dimension tables (Chapter 4)

3.4 Handling Missing and Null Values

Blank answers were replaced with named placeholders instead of being left null, so a DAX aggregation never fails on an empty cell and the answer still appears as its own category in a chart. Flag columns such as SumulCSATFlag, CategoryCSATFlag, ButterBreadAwareFlag and ProbioticAwareFlag hold 1 or 0 so they can be summed or averaged directly.

3.5 Removing Duplicates

Each table is built at a fixed grain, so duplicates are prevented by design: one row per ResponseID in FactSurveyResponse, one row per rated attribute in FactAttributeRating and one row per product in FactProductPricePerception. [If you also ran Remove Duplicates in Power Query, add the number of rows removed here.]

3.6 Correcting Data Types and Formats

Every table has a Change Type step in Power Query. Keys such as CustomerKey, DateKey and LocationKey are whole numbers, DailyPouchMilkLtr and DailyLooseMilkLtr are decimals, EstimatedMonthlySpendINR is a whole number, Date and YearMonth are dates, and all category and comment columns are text. The rating columns (RatingPriceNum, RatingTasteNum, RatingPackagingNum, RatingAvailabilityNum, RatingSchemesNum) are whole numbers from 1 to 4.

3.7 Standardizing Categories and Column Names

Column names follow one pattern (PascalCase, no spaces) so that measures read clearly. Two calculated fields were engineered:

- TotalDailyMilkLtr = DailyPouchMilkLtr + DailyLooseMilkLtr.
- ConsumptionTier groups households as Light (under 2 L per day), Medium (2 to 4 L) or Heavy (over 4 L).

Each table is loaded with the same three Power Query steps. Figure 3.1 shows them for DimCustomer.

```
let
    Source = Csv.Document(File.Contents("...\04_Star_Schema\DimCustomer.csv"),
        [Delimiter=",", Columns=10, Encoding=1252, QuoteStyle=QuoteStyle.None]),
    #"Use First Row as Headers" = Table.PromoteHeaders(Source,
[PromoteAllScalars=true]),
    #"Change Type" = Table.TransformColumnTypes(#"Use First Row as Headers",
        {{"CustomerKey", Int64.Type}, {"CustomerID", type text}, {"CustomerName",
type                                                                    text},
         {"Gender", type text}, {"Age", Int64.Type}, {"AgeGroup", type text},
         {"HouseholdSize", Int64.Type}, {"CustomerSegment", type text},
         {"Email", type text}, {"ContactNo", Int64.Type}})
in
    #"Change Type"
```

Figure 3.1: Power Query steps (Source, Use First Row as Headers, Change Type) for DimCustomer. [Insert your Power Query Applied Steps screenshot here.]

3.8 Data Validation and Final Dataset

Row counts and totals were reconciled before the dashboard was built (Table 3.3).

Check	Expected	Result
Attribute ratings = responses × 5 attributes	36,750	36,750 rows in FactAttributeRating
Price ratings = responses × 7 products	51,450	51,450 rows in FactProductPricePerception
Distinct customers on the dashboard	4,028	4,028 rows in DimCustomer
Consumption tiers add up to total responses	7,350	4,840 + 2,022 + 488 = 7,350
Feedback sentiment shares add up to 100%	100%	81.56% + 4.52% + 13.92% = 100.00%

Chapter 4: Data Modeling Using Star Schema

4.1 Importance of Data Modeling

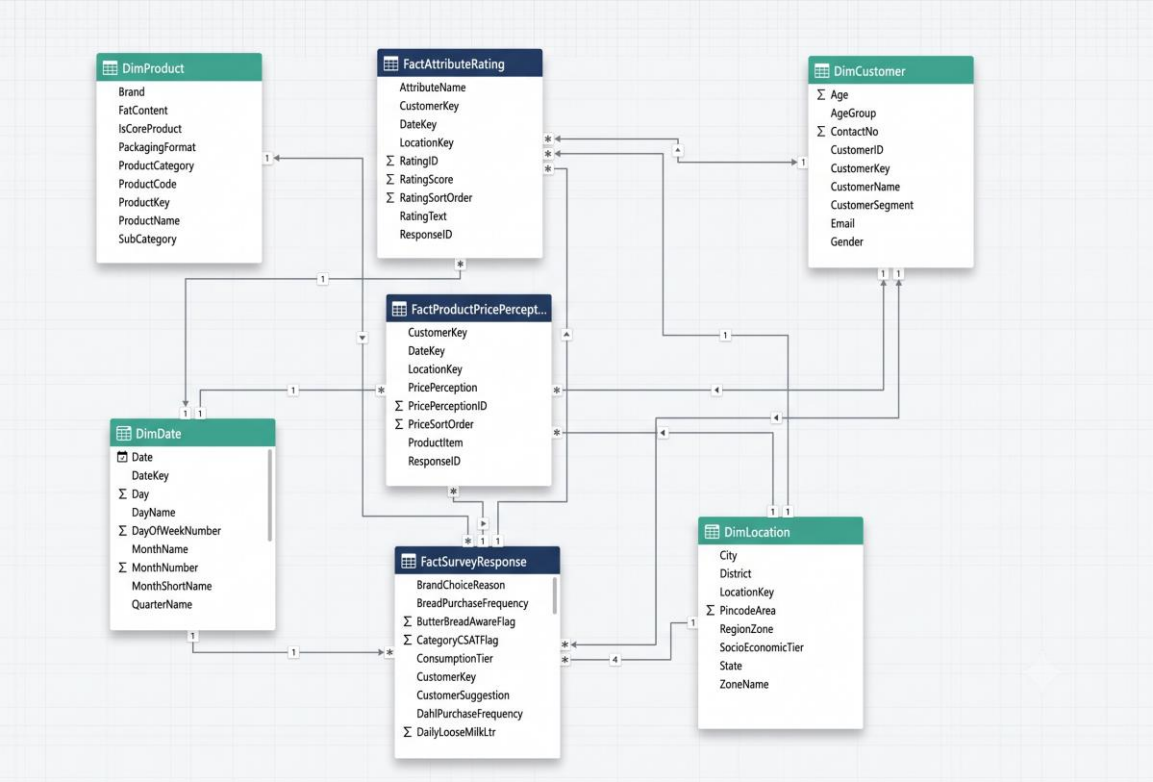
A flat survey spreadsheet forces demographics, ratings and answers into one wide table, which repeats values, uses more memory and makes some calculations ambiguous. A proper model keeps each kind of data in its own table and lets slicers such as Zone, Month and Customer Segment filter every visual in the same way, so the numbers agree from page to page.

4.2 Understanding Fact and Dimension Tables

A fact table holds measurable events: here the survey answers and ratings. A dimension table holds the descriptions that give those events context: who answered, where, when and about which product. Facts connect to dimensions through key columns (CustomerKey, DateKey, LocationKey, ProductKey).

4.3 Design of the Star Schema

The model is a multi-fact star schema, also called a galaxy schema. Three fact tables share the same conformed dimensions, so a single slicer such as Zone filters survey answers, attribute ratings and price perception together. Figure 4.1 shows the design and Figure 4.2 shows the same model in Power BI Model view.



The model separates three levels of detail. FactSurveyResponse holds one row per respondent. FactAttributeRating holds five rows per respondent (one for each rated attribute) and FactProductPricePerception holds seven (one for each product), so ratings and price opinions can be analysed by attribute and by product without crowding the main table.

4.4 Fact Table Structure

Fact table	Grain	Rows	Keys to dimensions	Main fields
FactSurveyResponse	One row per completed survey	7,350	CustomerKey, DateKey, LocationKey, ProductKey	Litres and spend, consumption tier, satisfaction and awareness flags,

Fact table	Grain	Rows	Keys to dimensions	Main fields
				brand and pack choice, bread and dahi frequency, feedback sentiment and suggestion
FactAttributeRating	One row per rated attribute per survey	36,750	CustomerKey, DateKey, LocationKey	AttributeName (availability, taste, packaging, price, schemes), RatingScore (1 to 4), RatingText, RatingSortOrder
FactProductPricePerception	One row per product per survey	51,450	CustomerKey, DateKey, LocationKey	ProductItem, PricePerception (low, reasonable, high), PriceSortOrder

FactSurveyResponse is the widest table (42 columns). Its columns fall into these groups:

Field group	Columns
Keys	ResponseID, DateKey, CustomerKey, ProductKey, LocationKey
Volume and spend	DailyPouchMilkLtr, DailyLooseMilkLtr, TotalDailyMilkLtr, EstimatedMonthlySpendINR, ConsumptionTier, MonthlySpendBracket
Ratings (1 to 4)	RatingPriceNum, RatingTasteNum, RatingPackagingNum, RatingAvailabilityNum, RatingSchemesNum, PouchOverallRatingNum, QualityOpinionNum, QualityOpinion
Flags (1 or 0)	SumulCSATFlag, CategoryCSATFlag, ButterBreadAwareFlag, ProbioticAwareFlag
Behaviour and choice	MilkTypeConsumed, DeliveryChannel, PreferredBrand, BrandChoiceReason, PurchasedPackVariant, SatisfactionDriver
Price perception by product	PriceMilkPerception, PriceGheePerception, PricePaneerPerception, PriceDahiPerception, PriceShrikhandPerception, PriceLassiPerception, PriceFlavourMilkPerception

Field group	Columns
Bakery and dairy	BreadPurchaseFrequency, DahiPurchaseFrequency
Feedback	FeedbackSentiment, FeedbackTheme, ReasonBuyNotBuy, CustomerSuggestion

4.5 Dimension Tables and Their Attributes

Dimension	Key	Rows	Attributes	Used for
DimCustomer	CustomerKey	4,028	CustomerID, CustomerName, Gender, Age, AgeGroup, HouseholdSize, CustomerSegment, Email, ContactNo	Age Group and Customer Segment slicers, segment charts
DimDate	DateKey (YYYYMMDD)	365	Date, Day, DayName, DayOfWeekNumber, WeekNumber, MonthNumber, MonthName, MonthShortName, QuarterNumber, QuarterName, Year, YearMonth, YearQuarter, WeekendFlag	Month slicer, monthly CSAT trend
DimLocation	LocationKey	12	ZoneName, RegionZone, City, District, State, PincodeArea, SocioEconomicTier	Zone slicer, zone charts
DimProduct	ProductKey	10	ProductCode, Brand, ProductName, ProductCategory, SubCategory, FatContent, PackagingFormat, IsCoreProduct	Product Name slicer, brand and pack analysis

4.6 Relationships and Cardinality

The model has ten active relationships. Each is many-to-one from a fact table to a dimension, with a single filter direction (Table 4.2).

From (many side)	To (one side)	Cardinality	Filter direction	Status
FactSurveyResponse[CustomerKey]	DimCustomer[CustomerKey]	Many to one	Single	Active
FactSurveyResponse[DateKey]	DimDate[DateKey]	Many to one	Single	Active
FactSurveyResponse[LocationKey]	DimLocation[LocationKey]	Many to one	Single	Active
FactSurveyResponse[ProductKey]	DimProduct[ProductKey]	Many to one	Single	Active
FactAttributeRating[CustomerKey]	DimCustomer[CustomerKey]	Many to one	Single	Active
FactAttributeRating[DateKey]	DimDate[DateKey]	Many to one	Single	Active
FactAttributeRating[LocationKey]	DimLocation[LocationKey]	Many to one	Single	Active
FactProductPricePerception[CustomerKey]	DimCustomer[CustomerKey]	Many to one	Single	Active
FactProductPricePerception[DateKey]	DimDate[DateKey]	Many to one	Single	Active
FactProductPricePerception[LocationKey]	DimLocation[LocationKey]	Many to one	Single	Active
FactAttributeRating[ResponseID]	FactSurveyResponse[ResponseID]	Many to one	Single	Inactive
FactProductPricePerception[ResponseID]	FactSurveyResponse[ResponseID]	Many to one	Single	Inactive

FactProductPricePerception stores the product as text in ProductItem, so it is filtered through the shared customer, date and location dimensions rather than through DimProduct.

4.7 Filter Direction and Relationship Validation

All active relationships filter from the dimension down to the facts. This avoids filter loops and ambiguous paths between the three fact tables. The two ResponseID links between facts are kept inactive for the same reason: they exist for drill-through and cross-referencing, and a measure can

switch one on with USERELATIONSHIP when it needs it. Each relationship was validated by selecting a dimension value (for example one zone) and checking that all three fact tables filtered together and that the totals matched the unfiltered counts of 7,350, 36,750 and 51,450 rows.

4.8 Benefits of the Final Data Model

- One zone, month or segment slicer filters survey answers, attribute ratings and price perception together.
- Ratings and price opinions are stored in narrow fact tables, so they can be charted by attribute and product without reshaping.
- Dimension tables are small (10 to 4,028 rows), so the model stays light and visuals respond quickly.
- A new zone, product or customer is added once to a dimension table, not to every fact row.

Chapter 5: Important Measures and DAX Calculations

5.1 Introduction to DAX Measures

Every number on the dashboard is a DAX measure, not a stored value, so it recalculates under the current slicers. The report holds 63 measures, kept in measure-only tables grouped by dashboard page (Measure, Measure-page3, Measure_Page_4, Measure-page_5,6 and NewMEASURES). The formulas of the main measures are in Appendix A.

5.2 Key Performance Indicators (KPIs)

Table 5.1 lists the measures behind the KPI cards, with the value each shows and what it means.

Group	Measure	Result shown	Business interpretation
Volume	Total Responses	7,350	Completed surveys: a count of rows in FactSurveyResponse
Volume	Distinct Customers	4,028	Unique customers, counted on CustomerKey
Satisfaction	Sumul Brand CSAT %	86.1%	Average of SumulCSATFlag: the share of customers satisfied with Sumul
Satisfaction	Category Milk CSAT %	91.7%	Average of CategoryCSATFlag: satisfaction with the milk category
Satisfaction	CSAT Target %	90%	Fixed target drawn on the gauge

Group	Measure	Result shown	Business interpretation
Consumption	Avg Daily Volume	2.49 L	Average of pouch plus loose litres per household
Consumption	Median Daily Volume	2.50 L	Middle household; close to the average, so few extreme values
Consumption	Avg Monthly Spend	₹4,519	Average of EstimatedMonthlySpendINR
Consumption	Pouch Milk Penetration %	72.4%	Responses where MilkTypeConsumed is pouch milk
Brand	Sumul Market Share %	77.8%	Share of responses preferring Sumul or Amul
Brand	Mahi Share %	11.5%	Share preferring Mahi, the closest competitor
Brand	Doorstep Delivery Share %	59.4%	Doorstep orders as a share of doorstep plus retailer orders
Brand	Amul Gold Share %	49.5%	Share of responses buying the Amul Gold pack
Quality	Pouch Overall Score Display	3.17 / 4.0	Average overall pouch rating, shown as a score out of 4
Quality	Net Quality Top Box %	70.75%	Share of quality opinions scoring 3 or more
Quality	Price Value Score Display	2.59 / 4.0	Average price rating
Quality	Promotinal Schemes Score Display	1.86 / 4.0	Average schemes rating: the lowest attribute
Bakery and dairy	Bread Buyer %	67.74%	Daily or twice-weekly bread buyers
Bakery and dairy	Bread Butter Awareness %	51.59%	Customers aware of butter bread

Group	Measure	Result shown	Business interpretation
Bakery and dairy	Dahi Buyer %	69.50%	Daily or twice-weekly dahi buyers
Bakery and dairy	Probiotic Dahi Awareness %	38.31%	Customers aware of probiotic dahi
Feedback	Positive, Neutral and Negative Feedback %	81.56%, 4.52%, 13.92%	Each sentiment as a share of all feedback
Feedback	Dissatisfaction Factor %	36.19%	Share of the five attribute ratings that are 1 or 2

5.3 Sales and Revenue Measures

The survey has no sales or revenue data, so there are no revenue measures. The closest is Avg Monthly Spend, which is the spend each customer reported, not SUMUL's sales.

5.4 Quantity and Volume Measures

Avg Pouch Milk Litres and Avg Loose Milk Litres average the two volume columns, and Avg Daily Volume and Median Daily Volume add them per household with AVERAGEX and MEDIANX. COALESCE treats a blank as zero so a household that buys only pouch milk is still counted. The consumption tiers (Light, Medium, Heavy) come from the ConsumptionTier column.

5.5 Customer or Product Performance Measures

Product performance is measured by attribute averages (Availability_Avg 3.30, Taste_Avg 3.27, Packaging_Avg 3.11, Price_Avg 2.59, Schemes_Avg 1.86) and by pack and brand shares. Display measures such as Pouch Overall Score Display join the average and the scale maximum into a text label like 3.17 / 4.0 for the KPI cards.

5.6 Survey-Based Measures

Survey measures count answers: Satisfied Responses and Not Satisfied Responses from the CSAT flag, Driver Responses and Driver Share % for the primary satisfaction driver, Category Satisfied % and Category Not Satisfied %, awareness flags and the bread and dahi frequency shares. Negative Theme Responses counts responses with negative feedback.

5.7 Percentage and Comparative Measures

Percentages use DIVIDE with a zero alternate result, so an empty filter shows 0 instead of an error. Shares such as Bread Frequency % and Negative Feedback Share % divide by the same count with REMOVEFILTERS on the column being charted, so each bar is a share of the whole. Sumul CSAT Prev Month uses DATEADD to compare each month with the one before. Respondent % of Total uses ALLSELECTED so a share is measured against the responses in the current selection.

5.8 Filter Context and Measure Validation

Each measure was tested with no filter (7,350 responses), with one slicer value and with two slicers combined. Totals were reconciled against the data: consumption tiers add up to 7,350 and the three

sentiment shares add up to 100%. Measures that must ignore a slicer, such as a share of the total, use REMOVEFILTERS or ALLSELECTED.

Chapter 6: Survey Questions and Insights

6.1 Objective of the Survey Analysis

The survey analysis answers five questions for the Marketing Department: how satisfied customers are, what drives that satisfaction, which brands and packs they choose, how they rate quality and price, and what they complain about.

6.2 Survey Dataset Overview

The survey has 7,350 responses from 4,028 distinct customers, collected across the Surat zones and grouped by month in 2025. Most percentages in this chapter are shares of the responses in the current filter.

6.3 Survey Questions and Response Options

The questionnaire, titled "Customer Behaviour and Satisfaction of Sumul Brand", has 20 numbered questions on three pages. The form numbers them 1, 2, 4 and so on, because it has no question 3; this report keeps the form's numbers. Table 6.1 lists every question with its response options. Two (Q10 and Q11) are open-ended, Q12 and Q14 are rating grids, and the rest are closed questions. The full wording is in Appendix B.

The dashboard groups some answers rather than showing them raw. Q5 litres are grouped into light (under 2 L), medium (2 to 4 L) and heavy (over 4 L) consumption tiers. Q12 ratings are scored 1 to 4 (Poor 1, Excellent 4). Q10 and Q11 are coded into a sentiment and a theme in the feedback analysis.

Q	Question	Response options
1	Which milk do you consume?	Pouch milk, loose milk, both
2	Are you satisfied with the milk you are consuming?	Yes, no
4	Do you get milk pouch at the doorstep or from a retailer?	Doorstep, from retailer
5	Total consumption of milk in a day	Pouch milk in litres, loose milk in litres (written in)
6	Your monthly expenditure on milk (₹)	1,000-1,500; 1,501-2,000; 2,001-2,500; above 2,500
7	In pouch milk, which brand do you prefer?	Sumul/Amul, Rajhansh, Mahi, Gokulam, other
8	Why do you prefer this brand?	Quality, price, taste, availability
9	If you buy Sumul/Amul milk, which pack do you purchase?	Amul Gold, Amul Taaza, Amul Slim & Trim, Sumul Cow Milk, other

Q	Question	Response options
10	What is the reason for buying or not buying Sumul?	Open-ended
11	What are your suggestions or expectations from Sumul?	Open-ended
12	How do you scale your pouch milk?	Price, taste, packaging, availability, schemes, each rated Poor, Average, Good or Excellent
13	Your opinion about the quality of Sumul Dairy products	Excellent, average, good, very good
14	Rank Sumul products in terms of price	Milk, ghee, paneer, dahi, shrikhand, lassi, flavoured milk, each marked Highly priced, Reasonable or Low priced
15	Are you satisfied with the milk and milk products of Sumul Dairy?	Yes, no
16	If yes, the reason for the satisfaction	Easy availability, density of milk, creaminess, freshness, hygiene, taste, price, any other
17	How frequently do you buy bread?	Daily, twice in a week, twice in a month, rarely
18	Do you know Amul has started manufacturing butter bread?	Yes, no
19	How frequently do you buy dahi?	Daily, twice in a week, twice in a month, rarely
20	Do you know about probiotic dahi?	Yes, no

6.4 Respondent Demographics

The top of the questionnaire records the respondent's name, address, age, gender and contact number. In the dashboard, age is grouped into 18-25 (Youth), 26-35 (Young Adults), 36-50 (Mature Families) and 51+ (Seniors). The address gives the zone: Adajan, Vesu, Athwa, Majura, Palanpur, City Light, Limbayat, Varachha, Rander, Udhna, Katargam and Amroli. Customer segments (Large Joint Family, Senior Citizen Household, Young Couple or Single, and Nuclear Family with children) are a household classification in the data model rather than a question on the form. Name and contact number are personal details and are not used in any analysis in this report.

6.5 Analysis of Individual Survey Questions

- **Satisfaction. (Q2 and Q15)** Sumul brand CSAT is 86.1% against a 90% target. Category CSAT is 91.7%, and every customer segment is above 91% (Table 6.2).

Customer segment	Category satisfied	Category not satisfied
Large joint family	92.41%	7.59%
Senior citizen household	91.98%	8.02%
Young couple or single	91.53%	8.47%
Nuclear family with children	91.43%	8.57%

- **Satisfaction drivers. (Q16)** Availability is the single biggest driver, and freshness plus hygiene together are larger than taste and texture (Table 6.3). The shares are percentages of all respondents and add to 86.26%, so the remaining respondents named no primary driver.

Driver group	Drivers included	Share of respondents
Availability	Easy availability	26.29%
Freshness and hygiene	Freshness 18.79%, hygiene 12.60%	31.39%
Taste and texture	Taste 11.56%, density 8.98%, creaminess 3.25%	23.79%
Price	Price	3.24%
Other	Any other	1.55%

- **Brand choice. (Q7 and Q8)** Buyers of each brand choose it for different reasons (Table 6.4).

Brand	Availability	Price	Quality	Taste
Sumul/Amul	22%	5%	56%	17%
Mahi	11%	20%	36%	32%
Gokulam	33%	44%	13%	10%
Rajhansh	28%	48%	15%	9%
Other	28%	41%	20%	11%

- **Quality and price. (Q12, Q13 and Q14)** The overall quality score is 3.17 out of 4. Availability (3.30) and taste (3.27) score highest, packaging scores 3.11, price 2.59 and promotional schemes 1.86. By quality opinion, 40.73% say Excellent and 28.45% Very good.
- **Bread and dahi. (Q17 to Q20)** 67.74% buy bread daily or twice a week, and 69.50% buy dahi that often. Butter bread awareness is 51.59% and probiotic dahi awareness is 38.31%.
- **Feedback. (Q10 and Q11)** 81.56% of feedback is positive, 4.52% neutral and 13.92% negative.

6.6 Response Distribution and Trends

Monthly Sumul CSAT moves in a narrow band of about 84% to 88% through 2025, with highs near 88% in February and September and the lowest month near 84% in April (read from the trend chart). It does not reach the 90% target in any month. Pouch milk is the format for 72.4% of customers, loose milk for 8.3% and both for 19.3%, and about two-thirds of customers (4,840 of 7,350) fall in the medium consumption tier.

6.7 Key Findings and Customer Preferences

- Customers choose Sumul/Amul mainly for quality (56% of its buyers); they choose Gokulam and Rajhansh mainly for price.
- Availability and freshness matter most to satisfaction, and price rarely is the main driver.
- Amul Gold is the backbone pack at 49.5% of pack contribution, followed by Amul Taaza at 21.43%.
- Doorstep delivery carries 59.4% of purchases and beats retailer pickup in every zone shown.
- Ghee (65%) and paneer (59%) are most often called highly priced; milk (69%) and dahi (75%) are mostly called reasonable.
- Probiotic dahi awareness ranges from 57.0% in Vesu to 28.5% in Amroli.

6.8 Marketing Implications of Survey Results

- Protect the availability and freshness that satisfy customers, and treat missed deliveries as the most damaging failure.
- Rework promotional schemes, the lowest-rated attribute, before cutting price.
- Target probiotic dahi and butter bread awareness in the zones and customer groups where it is lowest.
- Watch Mahi: its buyers weigh taste and price more than Sumul/Amul buyers do.

Chapter 7: Dashboard Design and Description

7.1 Dashboard Objectives

The dashboard lets a marketing manager see satisfaction against target on the first page, then move through customers, brands, product quality, bakery and dairy, and feedback, filtering by zone, month, segment or brand at any point.

7.2 Dashboard Layout and Design Principles

Every page uses the same frame so that users learn it once:

- A SUMUL-branded left panel with navigation buttons to the six pages.
- A title bar with the page name and the slicers for that page.
- A row of four KPI cards under the title.
- Four visuals below, two per row, each with a clear title.
- A consistent blue, teal and amber palette: blue for volume and brand, teal for positive results, amber for attention and red for problems.

7.3 Overview of Dashboard Pages

Page	Purpose	Key question answered
Executive Summary	Satisfaction against the 90% target	Are we meeting the satisfaction target?
Customer Profile	Who buys what and how much	How do customers consume and spend?
Brand Dynamics	Brand share, choice drivers, packs and delivery	Why do customers choose Sumul/Amul or a rival?
Product Analysis	Quality scores and price perception	Which attributes and products need work?
Dairy & Bakery	Bread and dahi habits and awareness	Where can value-added products grow?
Market Analysis	Sentiment, complaints and suggestions	What do unhappy customers say?
Home	Landing page with navigation to the six analysis pages	Where do I start?

7.4 Description of KPI Cards

Page	KPI cards
Executive Summary	Total Responses 7,350; Distinct Customers 4,028; Sumul Brand CSAT 86.1%; Category Milk CSAT 91.7%
Customer Profile	Avg Daily Volume 2.49; Median Daily Volume 2.50; Avg Monthly Spend 4,519; Pouch Milk

Page	KPI cards
	Penetration 72.4%
Brand Dynamics	Sumul/Amul Market Share 77.8%; Mahi Competitor Share 11.5%; Doorstep Delivery Share 59.4%; Amul Gold SKU Share 49.5%
Product Analysis	Pouch Overall Quality Score 3.17 out of 4; Net Quality Top-Box 70.75%; Price Value Score 2.59 out of 4; Promotional Schemes Score 1.86 out of 4
Dairy & Bakery	Regular Bread Buyers 67.74%; Bread Butter Awareness 51.59%; Regular Dahi Buyers 69.50%; Probiotic Dahi Awareness 38.31%
Market Analysis	Positive Feedback 81.56%; Neutral 4.52%; Negative 13.92%; Dissatisfaction Rate 36.19%

7.5 Description of Charts and Visuals

- **Gauge:** Sumul CSAT against the 90% target.
- **Bar and column charts:** satisfaction drivers, brand preference, consumption tiers, attribute scores, purchase frequency, awareness by zone and negative responses by brand.
- **Doughnut charts:** milk format split, pack variant contribution, quality opinion and feedback sentiment.
- **Line chart:** monthly CSAT trend for 2025.
- **100% stacked bars:** brand choice drivers, price perception by product, category CSAT by segment and butter bread awareness by frequency.
- **Matrix:** average monthly spend by age group and spend band.
- **Table:** Feedback Explorer listing zone, age group, brand, sentiment and each customer suggestion.

7.6 Slicers, Filters, and Interactivity

Page	Slicers
Executive Summary	Month, Zone, Customer Segment
Customer Profile	Month, Zone, Customer Segment
Brand Dynamics	Zone, Brand, Delivery Channel, Pack Variant
Product Analysis	Product Name, Quality Opinion, Zone
Dairy & Bakery	Age Group, Zone
Market Analysis	Sentiment, Feedback Theme, Zone

Slicers sit in the title bar, so the filter choice is always visible. Each slicer filters every visual on its page through the star schema.

7.7 Drill-Down and Cross-Filtering

Clicking a bar, slice or point in any visual filters the other visuals on the page, which is Power BI's default cross-filtering. Hovering over a mark shows its value. Page navigation uses the six buttons in the left panel.

7.8 Explanation of Each Dashboard Page

1. Home page and design artwork



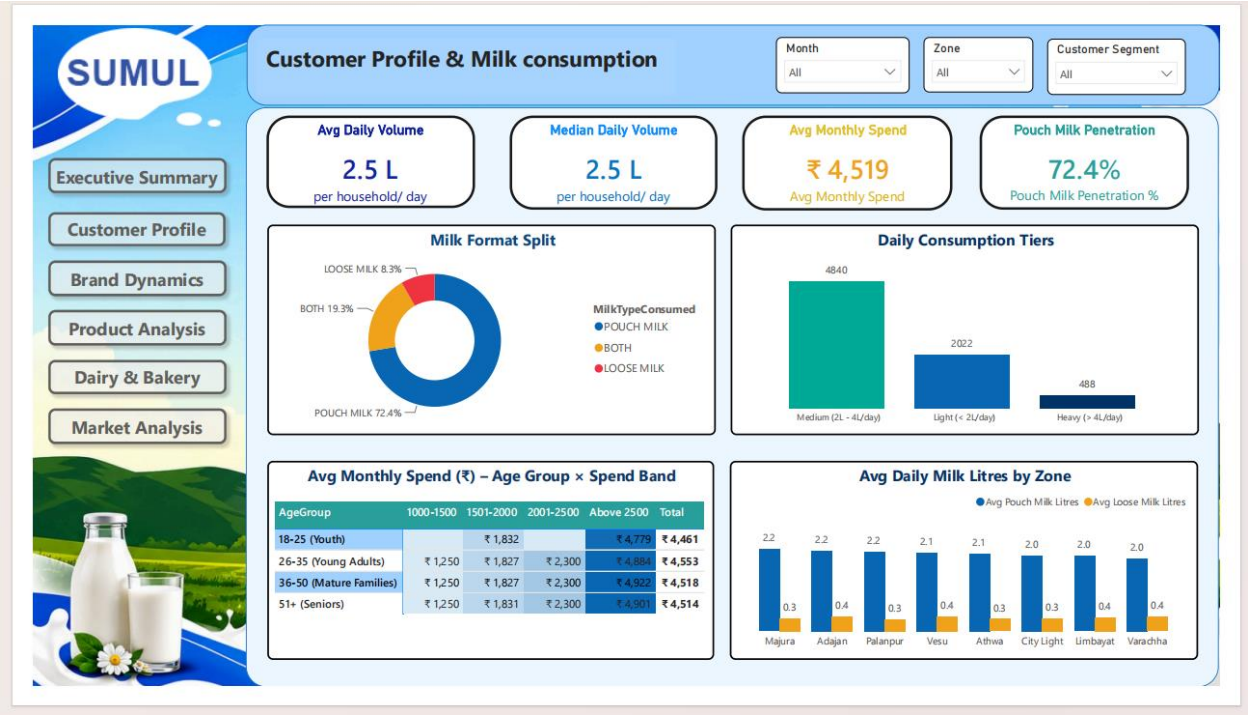
The Home page opens the report with the SUMUL dairy photograph (Figure 1.2) and buttons to the six analysis pages. Every page reuses the same milk-and-meadow artwork, SUMUL logo and navigation panel, which gives the report one consistent identity.

2. Executive Summary



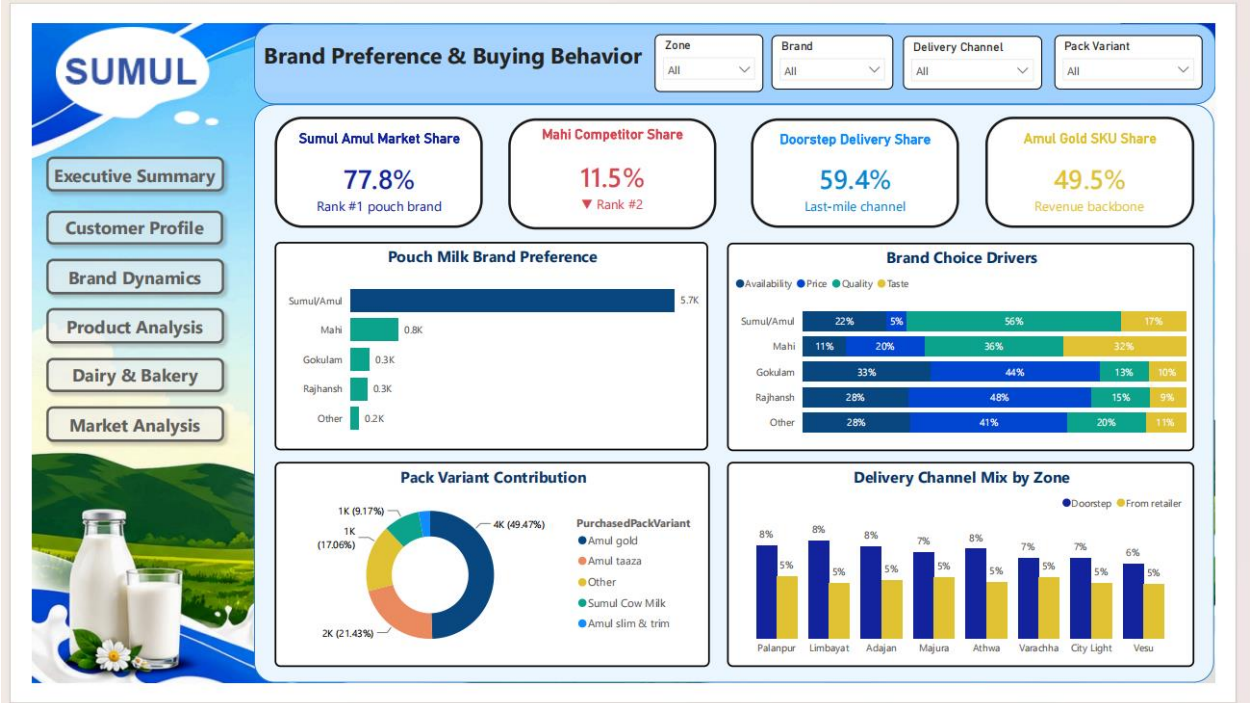
The gauge shows 86.1% against the 90% target. Easy availability (26.29%) and freshness (18.79%) lead the satisfaction drivers. Monthly CSAT stays around 84% to 88%, and all four customer segments rate the category above91%.

3. Customer Profile and Milk Consumption



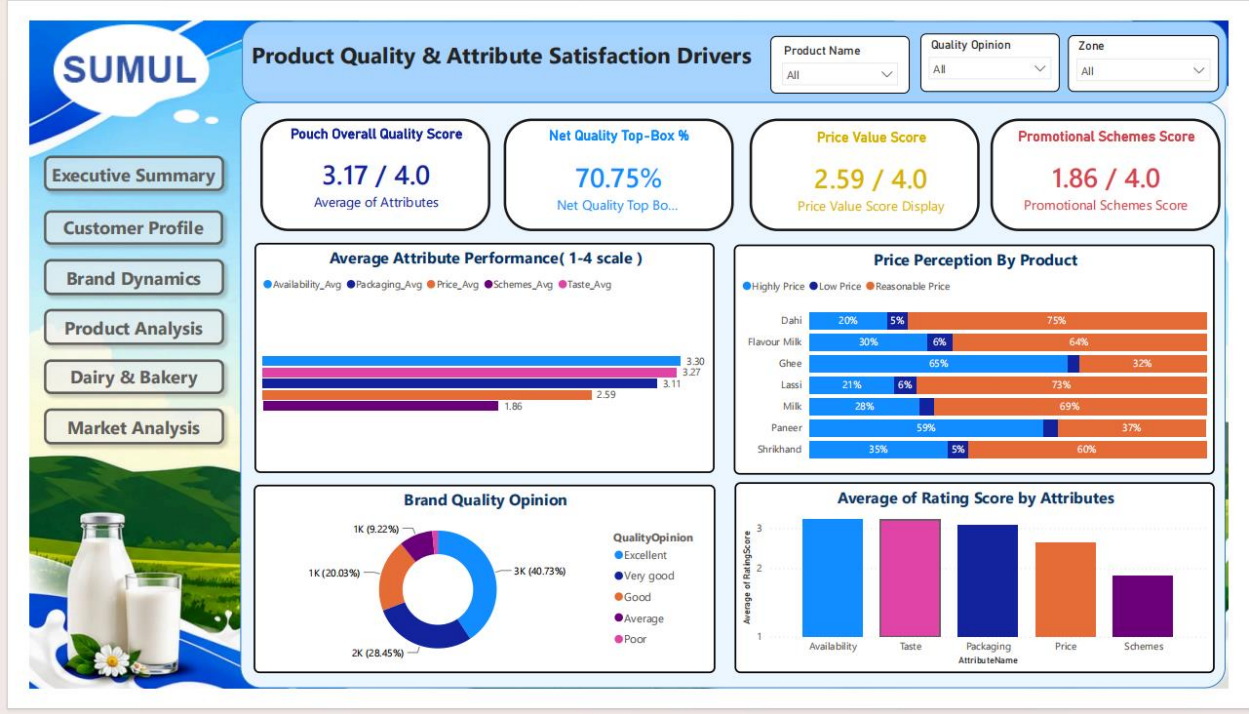
Pouch milk is 72.4% of customers, loose milk 8.3% and both 19.3%. Most customers are in the medium tier (4,840), then light (2,022) and heavy (488). Average spend is nearly the same for every age group, between ₹4,460 and ₹4,553, and pouch milk volume is 2.0 to 2.2 litres per day in every zone.

4. Brand Preference and Buying Behavior



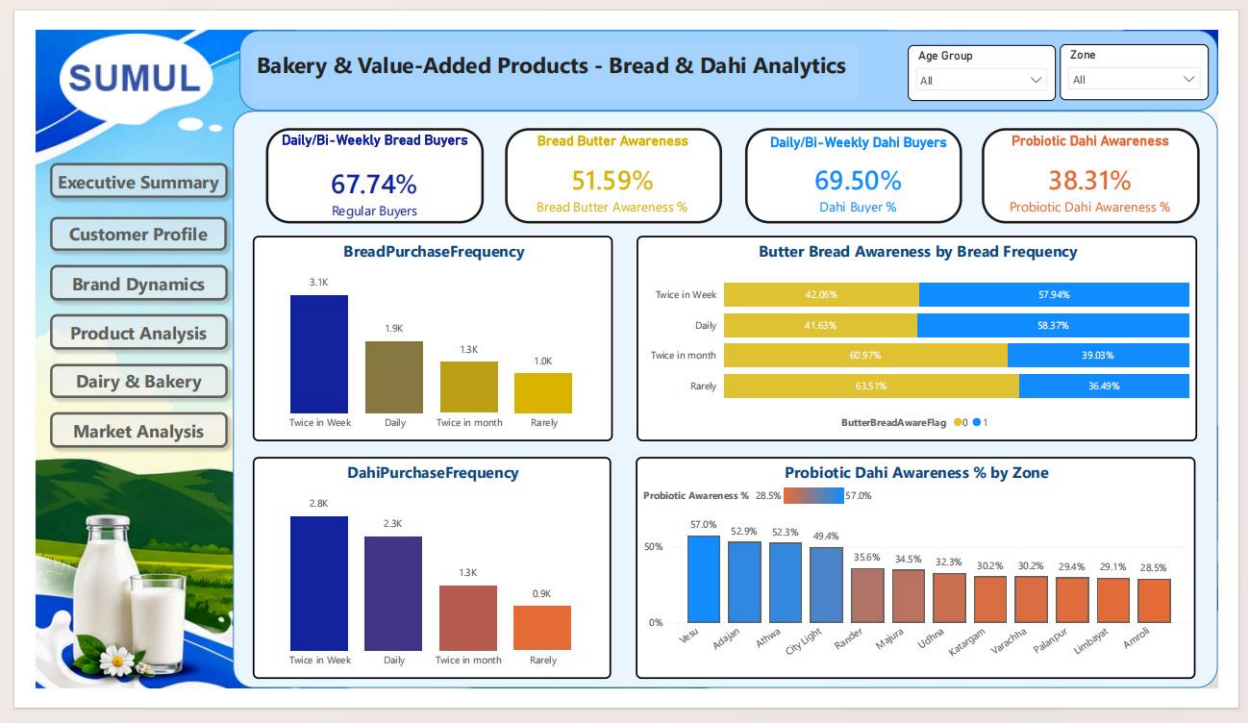
Sumul/Amul leads with about 5.7 thousand preferences against 0.8 thousand for Mahi. Sumul/Amul buyers cite quality (56%), while Gokulam and Rajhansh buyers cite price. Amul Gold is 49.47% of pack contribution, and doorstep delivery is higher than retailer pickup in every zone.

5. Product Quality and Attribute Satisfaction Drivers



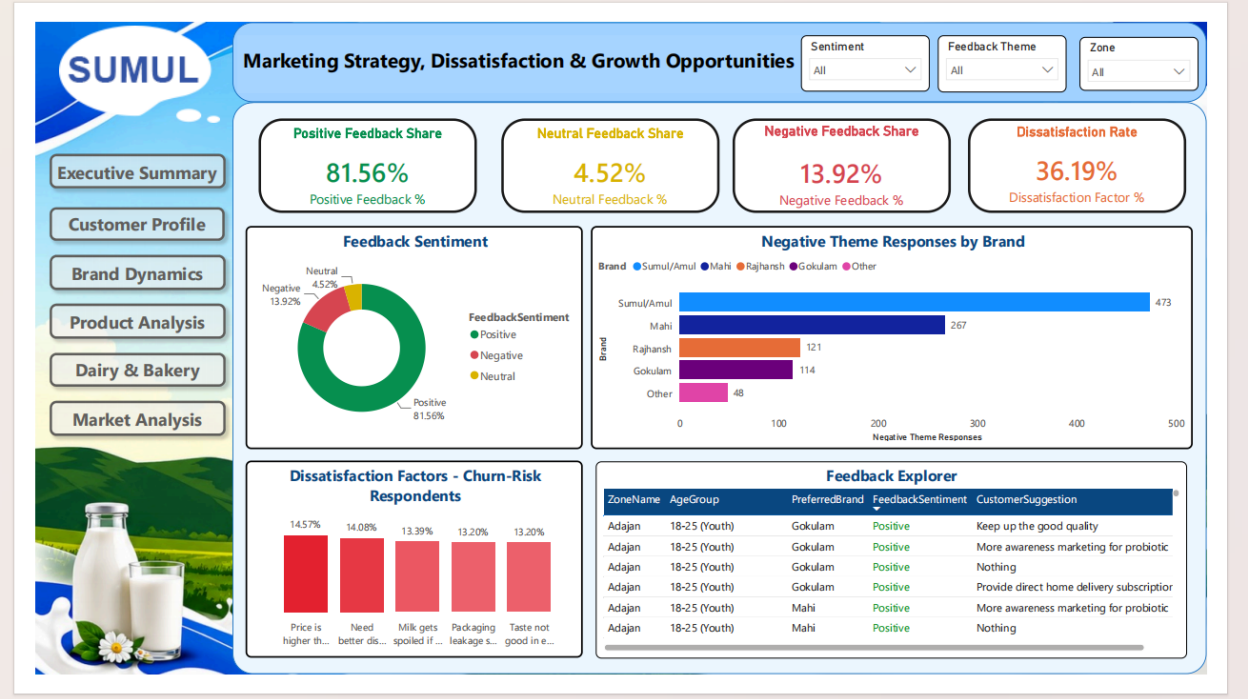
Availability scores 3.30 and taste 3.27, while price scores 2.59 and schemes 1.86. Ghee (65%) and paneer (59%) are most often seen as highly priced. The quality opinion doughnut shows 40.73% Excellent and 28.45% Very good.

6. Bakery and Value-Added Products



Bread is bought twice a week by about 3.1 thousand respondents and daily by about 1.9 thousand. Butter bread awareness is about 58% among daily and twice-weekly buyers but under 40% among occasional buyers. Probiotic dahi awareness is highest in Vesu (57.0%) and lowest in Amroli (28.5%).

7. Marketing Strategy, Dissatisfaction and Growth Opportunities



Negative responses are highest for Sumul/Amul (473) then Mahi (267). Among churn-risk respondents, the largest dissatisfaction factors are price and schemes (354) and packaging quality (345). The Feedback Explorer lists the written suggestions, for example improving packaging to prevent tears and adding morning delivery van trips.

7.9 How to Use the Dashboard

1. Open the Executive Summary and read the four KPI cards and the gauge.
2. Use Month, Zone and Customer Segment slicers to narrow to the group of interest.
3. Click a bar or slice to see how the other visuals change, and click it again to clear.
4. Use the left-panel buttons to move to the page that explains what you saw.
5. In the Feedback Explorer on Market Analysis, filter by sentiment, theme or zone to read the actual customer comments.

Chapter 8: Findings and Recommendations

8.1 Major Findings from the Dashboard

1. Sumul brand CSAT is 86.1%, 3.9 points below the 90% target and 5.6 points below category CSAT (91.7%).
2. Sumul/Amul leads pouch milk with 77.8% share; Mahi is second with 11.5%.
3. Availability and freshness are the top satisfaction drivers; price is rarely the main reason.
4. Promotional schemes (1.86 out of 4) and price value (2.59) are the weakest rated attributes.
5. Price and schemes (354) and packaging quality (345) make up about 68% of the 1,023 negative theme responses.
6. Probiotic dahi awareness is only 38.31%, with large gaps between zones.

8.2 Key Marketing Insights

- Sumul is not losing on product quality. Quality is rated 3.17 out of 4 and 69% rate the brand Excellent or Very good, so the gap to the category lies in experience factors such as packaging, schemes and delivery.
- Sumul/Amul has the most negative responses (473) but also the largest customer base. Mahi buyers are far more critical relative to size, because its 267 negative responses come from a much smaller base.
- Competitors win on price among price-led buyers (Gokulam 44%, Rajhansh 48%), a different battle from the quality-led Sumul/Amul base.

8.3 Customer and Product Trends

- Monthly CSAT moves between about 84% and 88% and has not reached 90% in any month of 2025.
- Pouch milk dominates every zone, and doorstep delivery leads retailer pickup in every zone shown.
- Amul Gold carries 49.47% of pack contribution, so the range depends on one SKU.
- Regular bread (67.74%) and dahi (69.50%) buying shows a strong base for value-added products.

8.4 Recommendations for the Marketing Department

Recommendation	Evidence from the dashboard	Expected benefit
Close the CSAT gap	Brand CSAT 86.1% against 90% target; availability and freshness are the top drivers	Reach the target by protecting the two things customers value most
Fix packaging quality	Packaging score 3.11 out of 4; 345 negative responses; customers report tearing	Fewer complaints and less churn risk
Rework promotional schemes	Schemes score 1.86 out of 4; price and schemes is the top dissatisfaction factor (354)	Better value perception without a blanket price cut
Strengthen doorstep delivery	Doorstep carries 59.4% of purchases; customers ask for	Higher availability, the strongest

Recommendation	Evidence from the dashboard	Expected benefit
	more morning van trips	satisfaction driver
Grow value-added awareness	Probiotic dahi awareness 38.31%, lowest in Amroli (28.5%) and Limbayat (29.1%); butter bread awareness under 40% among occasional buyers	Higher trial of dahi and butter bread
Watch Mahi	Mahi share 11.5%; its buyers weigh taste (32%) and price (20%) more than Sumul/Amul buyers (17% and 5%)	Early defence of share in contested zones
Keep evening stock reliable	The project analysis notes stock-outs at retail counters in the evening; confirm in the Feedback Explorer	Captures secondary daily purchases and improves availability, the top satisfaction driver

8.5 Potential Business Benefits

Acting on these findings could raise satisfaction towards the 90% target, reduce complaints about packaging and schemes, increase awareness of higher-margin products and keep Sumul/Amul's share lead over Mahi. The dashboard also lets the department check whether each action works by watching the same KPIs month after month.

Chapter 9: Conclusion and Future Scope

9.1 Project Summary

The project converted the SUMUL customer survey of 7,350 responses into a six-page Power BI dashboard. The data was cleaned in Power Query, modelled as a star schema, measured with DAX and presented with KPI cards, charts and slicers. The main result is that Sumul brand satisfaction (86.1%) is below the 90% target, and that packaging, promotional schemes and delivery are where customers are least satisfied.

9.2 Achievement of Project Objectives

Objective	Outcome
Clean and validate the survey data	Data profiled and cleaned in Power Query; totals reconcile across visuals (Chapter 3)
Build a star-schema model	Three fact tables and four dimension tables with many-to-one relationships (Chapter 4)
Create DAX measures	Measures for satisfaction, share, quality, awareness and sentiment (Chapter 5, Appendix A)
Analyse survey responses	Analysis by zone, age group, segment and brand (Chapter 6)
Design an interactive dashboard	Six pages with slicers and navigation (Chapter 7)
Recommend marketing actions	Six evidence-based recommendations (Chapter 8)

9.3 Skills and Learning Outcomes

- Data cleaning and profiling with Power Query.
- Dimensional modelling with a star schema.
- Writing and validating DAX measures under filter context.
- Choosing the right chart for each business question.
- Turning survey results into marketing recommendations.

9.4 Project Limitations

- The survey has no sales or revenue data, so findings describe customer opinion, not SUMUL's sales.
- The dashboard covers one survey period in 2025 and cannot yet show year-on-year change.
- Free-text suggestions are grouped by theme but not analysed with text analytics.
- Results depend on who answered the survey, so zones with fewer responses should be read with care.

9.5 Future Enhancements

- Connect the dashboard to a live data source so it refreshes on its own.
- Add sales and distribution data to link satisfaction to revenue.
- Add text analytics to the customer suggestions.

- Add targets and alerts for each zone, not only the overall 90% target.
- Publish to the Power BI Service with row-level security so zone managers see their own zone.
- We can expand the survey data to cover more district other than Surat as well to know the product \brand outreach.

References

1. Microsoft Learn, Power BI documentation.
2. Microsoft Learn, Understand star schema and the importance for Power BI.
3. Microsoft Learn, DAX function reference.
4. Microsoft Learn, Power Query documentation.
5. SUMUL Dairy, company profile and product information [add source and date].
6. SUMUL customer survey data, Marketing Department, 2025.

Appendix A: Important DAX Measures

The formulas below are taken from the Power BI file. Purpose and business interpretation of each measure are in Table 5.1.

```
Total Responses = COUNTROWS ( FactSurveyResponse )

Distinct Customers = DISTINCTCOUNT ( FactSurveyResponse[CustomerKey] )

Sumul Brand CSAT % = AVERAGE ( FactSurveyResponse[SumulCSATFlag] )

Category Milk CSAT % = AVERAGE ( FactSurveyResponse[CategoryCSATFlag] )

CSAT Target % = 0.90

Sumul CSAT Prev Month =
CALCULATE ( [Sumul Brand CSAT %], DATEADD ( DimDate[Date], -1, MONTH ) )

Driver Share % =
VAR CurrentDriverCount = [Driver Count Test]
VAR AllDriverCount =
    SUMX ( ALL ( FactSurveyResponse[SatisfactionDriver] ), CALCULATE ( [Driver
Count Test] ) )
RETURN DIVIDE ( CurrentDriverCount, AllDriverCount, 0 )

Avg Monthly Spend = AVERAGE ( FactSurveyResponse[EstimatedMonthlySpendINR] )

Avg Daily Volume =
AVERAGEX (
    FactSurveyResponse,
    COALESCE ( FactSurveyResponse[DailyPouchMilkLtr], 0 )
        + COALESCE ( FactSurveyResponse[DailyLooseMilkLtr], 0 )
)

Median Daily Volume =
MEDIANX (
    FactSurveyResponse,
    COALESCE ( FactSurveyResponse[DailyPouchMilkLtr], 0 )
        + COALESCE ( FactSurveyResponse[DailyLooseMilkLtr], 0 )
)

Pouch Milk Penetration % =
DIVIDE (
    CALCULATE (
        DISTINCTCOUNT ( FactSurveyResponse[ResponseID] ),
        FactSurveyResponse[MilkTypeConsumed] = "POUCH MILK"
    ),
    DISTINCTCOUNT ( FactSurveyResponse[ResponseID] ),
```

```
    0
)

Sumul Market Share % =
DIVIDE (
    CALCULATE (
        COUNTROWS ( 'FactSurveyResponse' ),
        'FactSurveyResponse'[PreferredBrand] IN { "Sumul", "Amul", "Sumul/Amul" }
    ),
    [Total Respondents],
    0
)

Mahi Share % =
DIVIDE (
    CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[PreferredBrand] = "Mahi" ),
    [Total Respondents],
    0
)

Doorstep Delivery Share % =
DIVIDE (
    CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[DeliveryChannel] = "Doorstep" ),
    CALCULATE (
        COUNTROWS ( 'FactSurveyResponse' ),
        'FactSurveyResponse'[DeliveryChannel] IN { "Doorstep", "From retailer" }
    ),
    0
)

Amul Gold Share % =
DIVIDE (
    CALCULATE (
        COUNTROWS ( 'FactSurveyResponse' ),
        'FactSurveyResponse'[PurchasedPackVariant] = "Amul gold"
    ),
    [Total Respondents],
    0
)

Respondent % of Total =
DIVIDE (
    COUNTROWS ( 'FactSurveyResponse' ),
    CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ), ALLSELECTED (
'FactSurveyResponse' ) ),
    0
)

Pouch Overall Score Display =
VAR AvgValue = AVERAGE ( FactSurveyResponse[PouchOverallRatingNum] )
VAR Maxvalue = MAX ( FactSurveyResponse[PouchOverallRatingNum] )
RETURN IF ( NOT ( ISBLANK ( AvgValue ) ), FORMAT ( AvgValue, "0.00" ) & " / " &
FORMAT ( Maxvalue, "0.0" ) )

Net Quality Top Box % =
VAR TotalResponses = COUNT ( FactSurveyResponse[QualityOpinionNum] )
VAR PositiveResponses =
```

```
    CALCULATE (
        COUNT ( FactSurveyResponse[QualityOpinionNum] ),
        FactSurveyResponse[QualityOpinionNum] >= 3
    )
RETURN DIVIDE ( PositiveResponses, TotalResponses, 0 )

Availability_Avg = AVERAGE ( FactSurveyResponse[RatingAvailabilityNum] )
Taste_Avg        = AVERAGE ( FactSurveyResponse[RatingTasteNum] )
Packaging_Avg    = AVERAGE ( FactSurveyResponse[RatingPackagingNum] )
Price_Avg        = AVERAGE ( FactSurveyResponse[RatingPriceNum] )
Schemes_Avg      = AVERAGE ( FactSurveyResponse[RatingSchemesNum] )

Bread Buyer % =
VAR _TargetBuyers =
    CALCULATE (
        COUNTROWS ( 'FactSurveyResponse' ),
        'FactSurveyResponse'[BreadPurchaseFrequency] IN { "Daily", "Twice in Week"
}
    )
VAR _TotalRespondents =
    CALCULATE (
        COUNTROWS ( 'FactSurveyResponse' ),
        REMOVEFILTERS ( 'FactSurveyResponse'[BreadPurchaseFrequency] )
    )
RETURN DIVIDE ( _TargetBuyers, _TotalRespondents, 0 )

Bread Butter Awareness % =
VAR _AwareCount =
    CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[ButterBreadAwareFlag] = 1 )
VAR _TotalCount = COUNTROWS ( 'FactSurveyResponse' )
RETURN DIVIDE ( _AwareCount, _TotalCount, 0 )

Probiotic Dahi Awareness % =
VAR _AwareCount =
    CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[ProbioticAwareFlag] = 1 )
VAR _TotalCount = COUNTROWS ( 'FactSurveyResponse' )
RETURN DIVIDE ( _AwareCount, _TotalCount, 0 )

Positive Feedback Share % =
DIVIDE (
    CALCULATE ( [Survey Responses], FactSurveyResponse[FeedbackSentiment] =
"Positive" ),
    CALCULATE ( [Survey Responses], REMOVEFILTERS (
FactSurveyResponse[FeedbackSentiment] ) ),
    0
)

Negative Theme Responses =
CALCULATE ( [Survey Responses], FactSurveyResponse[FeedbackSentiment] = "Negative"
)

Dissatisfaction Factor % =
VAR _TotalNegativeScores =
    CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[RatingPriceNum] IN { 1, 2 } )
  + CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[RatingSchemesNum] IN { 1, 2 } )
```

```
  + CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[RatingAvailabilityNum] IN { 1, 2 } )
  + CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[RatingTasteNum] IN { 1, 2 } )
  + CALCULATE ( COUNTROWS ( 'FactSurveyResponse' ),
'FactSurveyResponse'[RatingPackagingNum] IN { 1, 2 } )
-- five rating columns per response row
VAR _TotalPossibleScores = COUNTROWS ( 'FactSurveyResponse' ) * 5
RETURN DIVIDE ( _TotalNegativeScores, _TotalPossibleScores, 0 )
```

Supporting measures used above: Total Respondents = COUNTROWS (FactSurveyResponse); Survey Responses = DISTINCTCOUNT (FactSurveyResponse[ResponseID]); Driver Count Test counts responses where SatisfactionDriver is not blank.

Appendix B: Survey Questionnaire

Questionnaire: Customer Behaviour and Satisfaction of Sumul Brand. Respondent details: name, address, age, gender, contact number. The wording below follows the original form, including its numbering (there is no question 3).

Q1. Which milk do you consume? (a) Pouch milk (b) Loose milk (c) Both

Q2. Are you satisfied with the milk you are consuming? (a) Yes (b) No

Q4. Do you get milk pouch at: (a) Doorstep (b) From retailer

Q5. Total consumption of milk in a day: (a) Pouch milk in litres (b) Loose milk in litres

Q6. Your monthly expenditure on milk (in ₹): (a) 1000-1500 (b) 1501-2000 (c) 2001-2500 (d) Above 2500

Q7. In pouch milk, which brand do you prefer? (a) Sumul/Amul (b) Rajhansh (c) Mahi (d) Gokulam (e) Other

Q8. Why do you prefer this brand, due to: (a) Quality (b) Price (c) Taste (d) Availability

Q9. If you buy Sumul/Amul milk, which pack do you purchase? (a) Amul Gold (b) Amul Taaza (c) Amul Slim & Trim (d) Sumul Cow Milk (e) Other

Q10. What is the reason for buying or not buying Sumul? (open answer)

Q11. What are your suggestions or expectations from Sumul? (open answer)

Q12. How do you scale your pouch milk? Rate Price, Taste, Packaging, Availability and Schemes as Poor, Average, Good or Excellent.

Q13. What is your opinion about the quality of Sumul Dairy products? (a) Excellent (b) Average (c) Good (d) Very good

Q14. Give the rank of Sumul milk products in terms of price. For Milk, Ghee, Paneer, Dahi, Shrikhand, Lassi and Flavour milk, tick Highly priced, Reasonable price or Low price.

Q15. Are you satisfied with the milk and milk products of Sumul Dairy? (a) Yes (b) No

Q16. If yes, the reasons for the satisfaction: (a) Easy availability (b) Density of milk (c) Creaminess (d) Freshness (e) Hygiene (f) Taste (g) Price (h) Any other

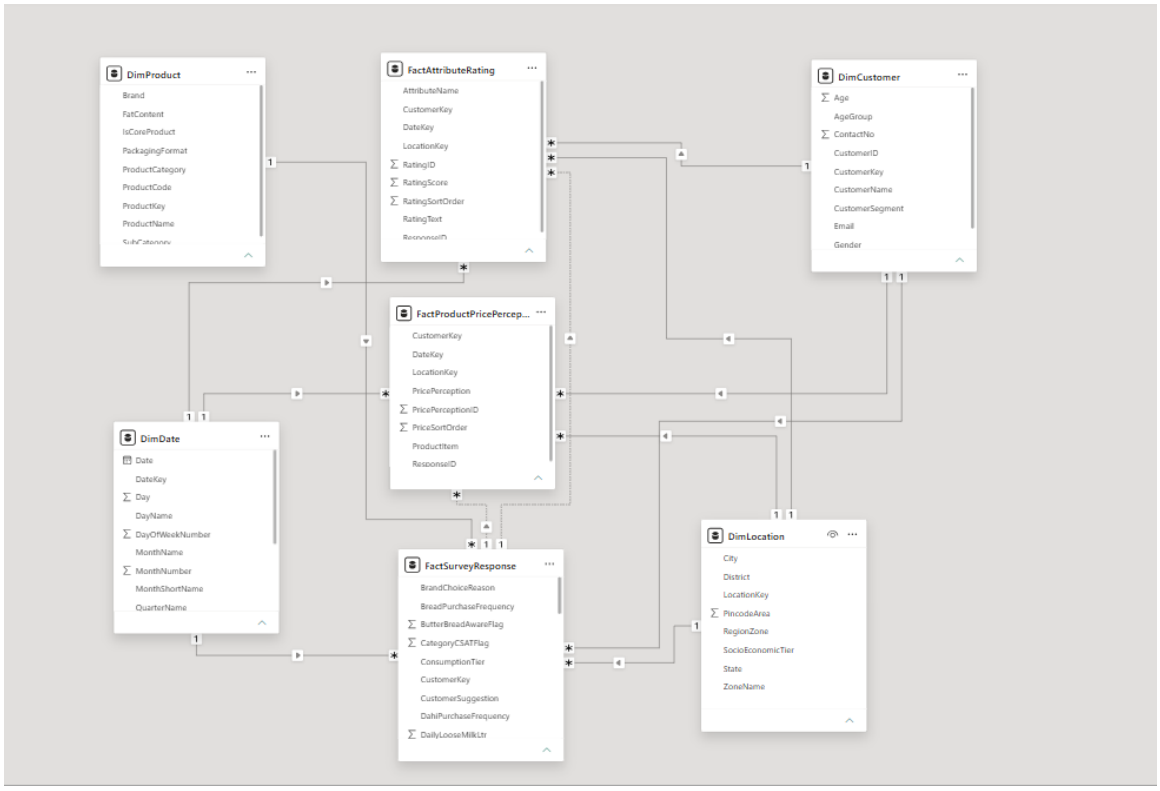
Q17. How frequently do you buy bread? (a) Daily (b) Twice in a week (c) Twice in a month (d) Rarely

Q18. Do you know Amul has started manufacturing butter bread? (a) Yes (b) No

Q19. How frequently do you buy dahi? (a) Daily (b) Twice in a week (c) Twice in a month (d) Rarely

Q20. Do you know about probiotic dahi? (a) Yes (b) No

Appendix C: Data Model Screenshot



Appendix D: Dashboard Screenshots

The full-page screenshots of all six dashboard pages are Figures 7.1 to 7.6 in Section 7.8.